

Point

From Number Theory to Algebraic Geometry and Logic

Session P3 — Algebraic Topology Crash Course

Complete lecture notes with proofs, examples, and exercises

Abstract

This session supplies the *topological prototype of Galois theory*. We build the fundamental group $\pi_1(X, x_0)$, the theory of covering spaces (lifting, monodromy, deck transformations, universal covers), and the classification of coverings by subgroups of π_1 — the exact categorical pattern that Session S35 reproduces for schemes as Grothendieck’s étale fundamental group, and that Act VII internalizes in topoi. The dictionary at the end (coverings $\leftrightarrow$ field extensions, deck groups $\leftrightarrow$ Galois groups, universal cover $\leftrightarrow$ separable closure) is the bridge the whole course walks across. Scope: no homology, no higher homotopy — just enough to make étale π_1 intuitive.

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: why coverings are the prototype of Galois theory
0:10–0:40	§1 Homotopy and π_1 : definitions, basepoint change, contractible, products
0:40–1:05	§2 Coverings: lifting, monodromy, deck transformations
1:05–1:15	<i>Break</i>
1:15–1:40	§2 cont.: universal covers; classification; $\pi_1(S^1) \cong \mathbb{Z}$
1:40–2:10	§3 The topological Galois correspondence; the dictionary; bridge to S16, S34, S35
2:10–2:30	Exercises / wrap-up: monodromy = evaluation along loops

Conventions

All spaces are path-connected and locally path-connected unless stated; maps are continuous. $[0, 1]$ carries the subspace topology of $\mathbb{R}$. Loops are paths with $\gamma(0) = \gamma(1) = x_0$. We write $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ and $T^2 = S^1 \times S^1$.

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1 Homotopy and the Fundamental Group

Definition 1.1 (Path; path homotopy). A **path** in X from a to b is a continuous $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = a$, $\gamma(1) = b$. A **path homotopy** (rel endpoints) between γ_0, γ_1 is a continuous $H : [0, 1] \times [0, 1] \rightarrow X$ with $H(s, 0) = \gamma_0(s)$, $H(s, 1) = \gamma_1(s)$, and $H(0, t) = a$, $H(1, t) = b$ for all t . We write $\gamma_0 \simeq \gamma_1$.

Definition 1.2 (Fundamental group). A **loop** at x_0 is a path with $a = b = x_0$. For loops γ, η define the concatenation

$$(\gamma * \eta)(s) = \begin{cases} \gamma(2s), & 0 \leq s \leq \frac{1}{2}, \\ \eta(2s - 1), & \frac{1}{2} \leq s \leq 1. \end{cases}$$

The **fundamental group** $\pi_1(X, x_0)$ is the set of path-homotopy classes $[\gamma]$ of loops at x_0 , with operation $[\gamma] \cdot [\eta] = [\gamma * \eta]$.

Proposition 1.3 (π_1 is a group). *The operation is well defined on homotopy classes and makes $\pi_1(X, x_0)$ a group: the identity is the class of the constant loop c_{x_0} , and the inverse of $[\gamma]$ is $[\bar{\gamma}]$ with $\bar{\gamma}(s) = \gamma(1 - s)$.*

Proof. If $\gamma_0 \simeq \gamma_1$ and $\eta_0 \simeq \eta_1$ via H, K , then gluing H and K at $s = \frac{1}{2}$ gives a path homotopy $\gamma_0 * \eta_0 \simeq \gamma_1 * \eta_1$: well-definedness. For the identity, given γ , the homotopy $H(s, t) = \gamma(\min\{1, 2s/(1+t)\})$? more transparently: reparametrize by a family of monotone maps $\varphi_t : [0, 1] \rightarrow [0, 1]$ with $\varphi_0(s) = \min\{2s, 1\}$ -type speed-ups; any two monotone reparametrizations fixing 0, 1 are homotopic through monotone maps, and precomposing γ with such a homotopy is a path homotopy. Applying this to the speed-ups that realize $\gamma * c$ and $c * \gamma$ and to $(\gamma * \eta) * \kappa$ vs $\gamma * (\eta * \kappa)$ gives identity and associativity up to path homotopy. For inverses, $H(s, t) = \gamma(t \cdot \min\{2s, 2 - 2s, 1\} + (1 - t) \cdot (\text{position}))$ contracts the backtrack $\gamma * \bar{\gamma}$ to c_{x_0} : explicitly $H(s, t) = \gamma((1 - t) \cdot \rho(s))$ where ρ traverses out-and-back; continuity is clear. $\square$

Proposition 1.4 (Basepoint change). *If α is a path from x_0 to x_1 , then $\beta_\alpha : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$, $[\gamma] \mapsto [\alpha^{-1} * \gamma * \alpha]$, is a well-defined group isomorphism, with inverse $\beta_{\alpha^{-1}}$. Hence for path-connected X , π_1 is independent of basepoint up to isomorphism.*

Proof. Concatenation respects path homotopy (Proposition 1.3), so β_α is well-defined; it is a homomorphism since the middle $\alpha * \alpha^{-1}$ cancels up to homotopy between products; $\beta_{\alpha^{-1}} \circ \beta_\alpha = [\alpha * (\alpha^{-1} * \gamma * \alpha) * \alpha^{-1}] = [\gamma]$ after cancelling, so they are mutually inverse. $\square$

Definition 1.5 (Homotopy equivalence; contractible). $f : X \rightarrow Y$ is a **homotopy equivalence** if $\exists g : Y \rightarrow X$ with $g \circ f \simeq \text{id}_X$ and $f \circ g \simeq \text{id}_Y$. X is **contractible** if $\text{id}_X \simeq$ a constant map.

Proposition 1.6. (a) *Homotopic maps $f_0, f_1 : X \rightarrow Y$ induce equal homomorphisms on fundamental groups (after adjusting basepoints along the track of the basepoint).*

(b) *A homotopy equivalence induces an isomorphism on π_1 .*

(c) *Contractible spaces have trivial π_1 at every basepoint.*

Proof ideas. (a) The homotopy H between f_0, f_1 sends a loop γ to a homotopy between $f_0 \circ \gamma$ and $f_1 \circ \gamma$ whose basepoint moves along $t \mapsto H(x_0, t)$; conjugating by that track (exactly Proposition 1.4) identifies the two induced maps. (b) Apply (a) to $g \circ f \simeq \text{id}$: $(g \circ f)_* = \text{id}_*$, and $(g \circ f)_* = g_* \circ f_*$, so f_* split-injects and similarly g_* , forcing isomorphisms. (c) A contractible X is homotopy equivalent to a point, whose π_1 is trivial. $\square$

Example 1.7 (First computations). • $\mathbb{R}^n$ (and any convex set) is contractible: $H(s, t) = (1 - t)s + tx_0$; hence $\pi_1 = 0$.

- **Products:** $\pi_1(X \times Y, (x, y)) \cong \pi_1(X, x) \times \pi_1(Y, y)$: the map $[\gamma] \mapsto ([p_X \circ \gamma], [p_Y \circ \gamma])$ is an isomorphism with inverse $([\alpha], [\beta]) \mapsto [s \mapsto (\alpha(s), \beta(s))]$; both well-defined on homotopy classes. Hence $\pi_1(T^2) \cong \mathbb{Z}^2$ and $\pi_1(T^n) \cong \mathbb{Z}^n$.
- S^n for $n \geq 2$: $\pi_1(S^n) = 0$ (mention: any loop misses a point up to homotopy, and $S^n \setminus \{\text{pt}\} \cong \mathbb{R}^n$ is contractible).
- S^1 : computed in Theorem 2.13 via coverings — the heart of this session.

Remark 1.8 (Functoriality). $(X, x_0) \mapsto \pi_1(X, x_0)$ and $f \mapsto f_*$ is a functor from pointed spaces to groups; Proposition 1.6 says it factors through the homotopy category. This “algebra extracted from space” is the pattern S35 repeats with étale covers in place of loops.

2 Covering Spaces: Lifting, Monodromy, Deck Groups

Definition 2.1 (Covering map). $p : E \rightarrow X$ is a **covering map** if it is continuous, surjective, and every $x \in X$ has an open neighbourhood U that is **evenly covered**: $p^{-1}(U) = \bigsqcup_{i \in I} V_i$ with each $p|_{V_i} : V_i \rightarrow U$ a homeomorphism. Then E is a **covering space** of X ; the fiber over x is $p^{-1}(x)$ (discrete, of constant cardinality if X connected).

Example 2.2 (Coverings we will use). • $p : \mathbb{R} \rightarrow S^1, t \mapsto e^{2\pi it}$; fiber over 1 is $\mathbb{Z}$.

- $p_n : S^1 \rightarrow S^1, z \mapsto z^n$ ($n \geq 1$); fiber over 1 is the n -th roots of unity.
- $q : \mathbb{R}^2 \rightarrow T^2 = \mathbb{R}^2/\mathbb{Z}^2$; fiber $\mathbb{Z}^2$.
- Trivial coverings $X \times F \rightarrow X$; disjoint unions of copies of X .

Proposition 2.3 (Unique lifting of paths). *Let $p : E \rightarrow X$ be a covering, $\gamma : [0, 1] \rightarrow X$ a path from x_0 , and $e_0 \in p^{-1}(x_0)$. There exists a unique path $\tilde{\gamma} : [0, 1] \rightarrow E$ with $p \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(0) = e_0$.*

Proof. Existence: by compactness of $[0, 1]$, subdivide $0 = s_0 < s_1 < \dots < s_k = 1$ so that each $\gamma([s_j, s_{j+1}])$ lies in an evenly covered U_j . Lift inductively: on $[s_0, s_1]$, γ lands in U_0 and e_0 lies in exactly one sheet V_0 over U_0 ; set $\tilde{\gamma} = (p|_{V_0})^{-1} \circ \gamma$ there; its endpoint picks the sheet over U_1 ; continue. *Uniqueness:* the set $A = \{s : \tilde{\gamma}_1 = \tilde{\gamma}_2 \text{ on } [0, s]\}$ is nonempty; it is open (on a small evenly covered window, two lifts landing in the same sheet must agree since $p|_V$ is injective) and closed (limits, by continuity and Hausdorffness of E); $[0, 1]$ connected forces $A = [0, 1]$. $\square$

Proposition 2.4 (Homotopy lifting). *Let $p : E \rightarrow X$ be a covering and $H : [0, 1] \times [0, 1] \rightarrow X$ a path homotopy rel endpoints. Given a lift $\tilde{\gamma}_0$ of $H(\cdot, 0)$, there is a unique lift $\tilde{H}$ of H extending $\tilde{\gamma}_0$; it is again a path homotopy rel endpoints, and $\tilde{H}(1, t)$ is constant in t .*

Proof idea. Subdivide the square into small squares each mapping into an evenly covered open; lift square by square, row by row, using Proposition 2.3 on edges and uniqueness on overlaps to glue. The endpoint statement follows because $t \mapsto \tilde{H}(1, t)$ is a continuous map from a connected space into the discrete fiber $p^{-1}(\gamma(1))$, hence constant. $\square$

Definition 2.5 (Monodromy action). For a covering $p : E \rightarrow X$ and $x_0 \in X$, define an action of $\pi_1(X, x_0)$ on the fiber $F = p^{-1}(x_0)$ by

$$[\gamma] \cdot e_0 = \tilde{\gamma}_{e_0}(1),$$

where $\tilde{\gamma}_{e_0}$ is the lift of γ starting at e_0 . This makes F a π_1 -**set**; the permutation representation $\pi_1(X, x_0) \rightarrow \text{Sym}(F)$ is the **monodromy** of the covering.

Proposition 2.6. *The monodromy action is well defined (independent of the representative of $[\gamma]$) and is indeed a group action; hence homotopic loops permute the fiber identically.*

Proof. Well-definedness is exactly Proposition 2.4 (the lifted homotopy has constant endpoint track). Action axioms: $[c_{x_0}] \cdot e_0 = e_0$ since the constant path lifts constantly; $[\gamma] * [\eta]$ lifts by first lifting η from e_0 then γ from $\tilde{\eta}(1)$, and concatenating these lifts is the lift of $\gamma * \eta$ (reparametrization), giving $([\gamma][\eta]) \cdot e_0 = [\gamma] \cdot ([\eta] \cdot e_0)$. $\square$

Example 2.7 (Monodromy computations). • $p_n : S^1 \rightarrow S^1, z \mapsto z^n$: the fiber over 1 is μ_n ; the generator $[\gamma]$ (one turn) acts as $e \mapsto e \cdot e^{2\pi i/n}$: a cyclic permutation of order n .

- $p : \mathbb{R} \rightarrow S^1$: fiber $\mathbb{Z}$; $[\gamma]$ acts by translation $m \mapsto m + 1$; monodromy $\pi_1(S^1) \rightarrow \text{Sym}(\mathbb{Z})$ has image the translation group $\cong \mathbb{Z}$.

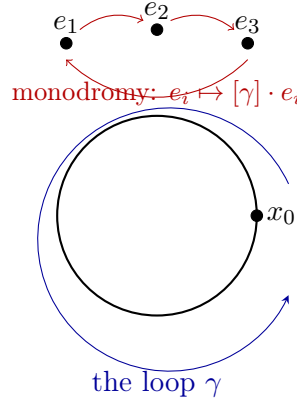


Figure 1: Monodromy: lifting the loop γ from each point of the fiber permutes the fiber. This permutation representation is the topological shadow of the Galois action on roots.

Definition 2.8 (Deck transformations). A **deck transformation** of $p : E \rightarrow X$ is a homeomorphism $\varphi : E \rightarrow E$ with $p \circ \varphi = p$; they form the **deck group** $\text{Deck}(p) = \text{Aut}(E/X)$.

Proposition 2.9. $\text{Deck}(p)$ acts freely on each fiber; if E is connected, a deck transformation is determined by its value at a single point; hence $|\text{Deck}(p)| \leq |p^{-1}(x)|$ for every x .

Proof. A deck transformation φ is a lift of $p : E \rightarrow X$ itself (since $p \circ \varphi = p$); two deck maps agreeing at one point are equal by the uniqueness argument of Proposition 2.3 applied to the identity path and connectedness of E . Freeness: if $\varphi(e) = e$ then $\varphi = \text{id}$ by the same uniqueness, so no nontrivial deck map fixes a fiber point. $\square$

Example 2.10. $\text{Deck}(\mathbb{R} \rightarrow S^1) = \{t \mapsto t + m : m \in \mathbb{Z}\} \cong \mathbb{Z}$; $\text{Deck}(p_n : S^1 \rightarrow S^1) = \{z \mapsto z\zeta : \zeta \in \mu_n\} \cong \mathbb{Z}/n$; $\text{Deck}(\mathbb{R}^2 \rightarrow T^2) \cong \mathbb{Z}^2$. In each case $|\text{Deck}|$ equals the fiber size: these coverings are *maximally symmetric*.

Definition 2.11 (Universal covering). A covering $p : \tilde{X} \rightarrow X$ with $\tilde{X}$ simply connected ($\pi_1 = 0$) is a **universal covering**. Examples: $\mathbb{R} \rightarrow S^1$, $\mathbb{R}^2 \rightarrow T^2$, $\mathbb{R}^n \rightarrow T^n$; existence holds for spaces that are path-connected, locally path-connected, and semi-locally simply connected.

Proposition 2.12 (Deck group of the universal cover). If $p : \tilde{X} \rightarrow X$ is a universal covering, then $\text{Deck}(p) \cong \pi_1(X, x_0)$; more precisely the monodromy action identifies the fiber $p^{-1}(x_0)$ with $\pi_1(X, x_0)$ and the deck action with left multiplication.

Proof idea. Since $\tilde{X}$ is simply connected, lifts of loops are unique and the endpoint map $[\gamma] \mapsto \tilde{\gamma}_{e_0}(1)$ is a bijection $\pi_1(X, x_0) \rightarrow p^{-1}(x_0)$ (surjectivity: join e_0 to any e by a path in $\tilde{X}$ and project). Each $g \in \pi_1$ then determines a unique deck map sending e_0 to $[\gamma] \cdot e_0$ (Proposition 2.9 plus lifting of the whole covering from the universal property), and composition matches group multiplication. $\square$

Theorem 2.13 ($\pi_1(S^1) \cong \mathbb{Z}$). *The map $\deg : [\gamma] \mapsto \tilde{\gamma}(1)$, where $\tilde{\gamma}$ is the lift of γ (based at $1 \in S^1$) starting at $0 \in \mathbb{R}$ along $p : \mathbb{R} \rightarrow S^1$, is a well-defined group isomorphism $\pi_1(S^1, 1) \xrightarrow{\cong} \mathbb{Z}$.*

Proof. *Well-defined:* $\tilde{\gamma}(1) \in p^{-1}(1) = \mathbb{Z}$; homotopic loops have lifts with equal endpoints (Proposition 2.4). *Homomorphism:* the lift of $\gamma * \eta$ is the lift of γ followed by the translate by $\deg(\gamma)$ of the lift of η (both are lifts of the second half with the same start), so $\deg(\gamma * \eta) = \deg(\gamma) + \deg(\eta)$. *Surjective:* $\gamma_n(t) = e^{2\pi i n t}$ lifts to $t \mapsto nt$, ending at n . *Injective:* if $\deg([\gamma]) = 0$ then $\tilde{\gamma}$ is a loop in $\mathbb{R}$ based at 0; $\mathbb{R}$ contractible gives a null-homotopy $\tilde{H}$; projecting, $p \circ \tilde{H}$ is a null-homotopy of γ rel basepoint. Hence $\deg$ is an isomorphism. $\square$

3 Classification of Coverings and the Galois Dictionary

Theorem 3.1 (Classification of coverings). *Let X be path-connected, locally path-connected, and semi-locally simply connected, with basepoint x_0 . The functor*

$$(E \xrightarrow{p} X) \longmapsto (p^{-1}(x_0), \text{monodromy})$$

is an equivalence between the category of coverings of X and the category of $\pi_1(X, x_0)$ -sets. Under it: connected coverings $\leftrightarrow$ transitive π_1 -sets; based connected coverings (up to based isomorphism) $\leftrightarrow$ subgroups of $\pi_1(X, x_0)$, via $H = p_\pi_1(E, e_0) = \{[\gamma] : [\gamma] \cdot e_0 = e_0\}$ (the stabilizer of e_0).*

Construction and proof idea. Inverse functor: given a π_1 -set S , form $E_S = (\tilde{X} \times S)/\pi_1$ with the diagonal action $[\gamma] \cdot (\tilde{x}, s) = (\tilde{x} \cdot [\gamma]^{-1}, [\gamma] \cdot s)$ using the universal cover $\tilde{X}$; the projection $E_S \rightarrow X$ is a covering with fiber S and the prescribed monodromy. Checking the two functors are mutually inverse uses Proposition 2.12 and the unique-lifting properties. The stabilizer description follows from the definition of monodromy: loops fixing e_0 are exactly those lifting to loops at e_0 , i.e. $p_*\pi_1(E, e_0)$. $\square$

Definition 3.2 (Normal (Galois) covering). A connected covering $p : E \rightarrow X$ is **normal** (or **Galois**) if $p_*\pi_1(E, e_0) \trianglelefteq \pi_1(X, x_0)$; equivalently $\text{Deck}(p)$ acts transitively on each fiber.

Proposition 3.3 (Topological Galois correspondence). *For X as above, with $\pi = \pi_1(X, x_0)$: connected coverings of X (up to equivalence) correspond to conjugacy classes of subgroups $H \leq \pi$; normal coverings to normal subgroups; and for the covering E_H corresponding to H ,*

$$\text{Deck}(E_H) \cong N_\pi(H)/H,$$

so in particular the universal cover (corresponding to $H = \{1\}$) has $\text{Deck} \cong \pi$, and a Galois covering has $\text{Deck} \cong \pi/H$.

Proof idea. From Theorem 3.1, $\text{Deck}(E_H)$ equals the π_1 -set automorphisms of the transitive set π/H , which are right multiplications by elements normalizing H , modulo those acting trivially (H itself): $N_\pi(H)/H$. Transitivity of the deck action $\iff$ normality of H is the same computation. $\square$

Example 3.4 (Coverings of S^1 classified). Subgroups of $\mathbb{Z}$ are $n\mathbb{Z}$; the connected covering for $n\mathbb{Z}$ is $p_n : S^1 \rightarrow S^1$, $z \mapsto z^n$ (for $n \geq 1$), with $\text{Deck} \cong \mathbb{Z}/n$; all are normal since $\mathbb{Z}$ is abelian; the universal cover is $n = 0$: $\mathbb{R} \rightarrow S^1$ with $\text{Deck} \cong \mathbb{Z}$. Compare: finite extensions of a field classified by subgroups of its Galois group in the finite case — the same shape.

Remark 3.5 (The dictionary: topology $\leftrightarrow$ Galois theory).

Topology (this session)	Galois theory (Acts II, VI; S35)
Covering $p : E \rightarrow X$	Field extension L/K
Connected / finite-sheeted covering	Field extension / finite extension
Universal cover $\tilde{X}$	Separable closure K^{sep}
Deck group $\text{Deck}(p)$	Galois group $\text{Gal}(L/K)$
Normal (Galois) covering	Galois extension
$\pi_1(X, x_0)$	Absolute Galois group G_K (S16)
Monodromy action on the fiber	Galois action on roots / on the fiber
Index $[\pi_1 : H]$	Degree $[L : K]$

Grothendieck's move (S35): for a scheme there is no classical topology, but the *category of finite étale covers* still exists; define $\pi_1^{\text{ét}}$ by demanding the same classification theorem (finite covers $\leftrightarrow$ finite continuous π_1 -sets). For $X = \text{Spec } K$ this recovers exactly G_K ; for nice complex varieties $\pi_1^{\text{ét}}$ is the profinite completion of the topological π_1 (Riemann existence theorem).

Remark 3.6 (Profinite nature; scope of this session). Because only *finite* covers are remembered algebraically, $\pi_1^{\text{ét}}$ is a **profinite group** (S34): an inverse limit of finite deck groups, e.g. $\pi_1^{\text{ét}}(\text{Spec } \mathbb{Q}) = G_{\mathbb{Q}} = \varprojlim \text{Gal}(L/\mathbb{Q})$. Monodromy of an infinite Galois extension is the inverse limit of finite permutation representations. Scope note: we deliberately stop here — no homology, no higher homotopy groups — because the course needs exactly this categorical pattern: *covers* $\leftrightarrow$ π_1 -sets.

Remark 3.7 (Monodromy as evaluation). Lifting a loop and watching the fiber permute is *analytic continuation of a point along a loop*: the fiber point is a “solution” (a root, a sheet) and the loop evaluates how solutions transform. This is the topological ancestor of “point = morphism” (S29, S54): a point of the fiber over x is a map into X together with a chosen lift, and monodromy is the action of loops on such choices.

4 Exercises

Exercise 1. Write out the explicit homotopies proving Proposition 1.3 (identity, associativity, inverse) without hand-waving reparametrizations.

Exercise 2. Prove Proposition 1.4 in full: well-definedness, homomorphism property, and $\beta_{\alpha^{-1}} = (\beta_{\alpha})^{-1}$. Show β_{α} depends only on the path-homotopy class of α .

Exercise 3. Prove $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$ carefully and deduce $\pi_1(T^n) \cong \mathbb{Z}^n$.

Exercise 4. Show that a disk D^2 , a cone, and $\mathbb{R}^n$ are contractible; conclude they are simply connected. Using Theorem 2.13, prove S^1 is *not* contractible.

Exercise 5. For $p_n : S^1 \rightarrow S^1$, $z \mapsto z^n$: describe the fiber over 1, the monodromy permutation of a generator, and $\text{Deck}(p_n)$; verify $|\text{Deck}| = n = \text{number of sheets}$.

Exercise 6. Prove Proposition 2.3 in full detail, including the Lebesgue-number subdivision for existence and the open-closed argument for uniqueness.

Exercise 7. For $p : \mathbb{R} \rightarrow S^1$: prove $\text{Deck}(p) \cong \mathbb{Z}$ explicitly and identify the monodromy homomorphism $\pi_1(S^1) \rightarrow \text{Deck}(p)$ under the isomorphism of Theorem 2.13.

Exercise 8. Reconstruct Theorem 2.13 from scratch using only the three lifting properties (path, homotopy, uniqueness): define $\deg$, prove it is a well-defined bijective homomorphism.

Exercise 9. Classify connected coverings of S^1 up to equivalence of coverings; match each with a subgroup of $\mathbb{Z}$; decide which are normal and compute their deck groups via Proposition 3.3.

Exercise 10. Show that a map homotopic to a constant induces the trivial homomorphism on π_1 ; deduce that π_1 cannot distinguish a point from $\mathbb{R}^n$ (motivating homology and higher invariants, not covered here).

Exercise 11 (The dictionary at work). Translate into covering language: (i) the quadratic extension $\mathbb{Q}(\sqrt{d})/\mathbb{Q}$; (ii) the cyclotomic extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$; (iii) the universal cover of S^1 . Then translate back: the covering $p_2 : S^1 \rightarrow S^1$; the trivial covering $X \times \{1, 2\} \rightarrow X$.

Exercise 12 (Challenge). State (without proof) that $\pi_1(S^1 \vee S^1)$ is free on two generators; describe its universal covering as a graph (the Cayley graph of F_2) and explain why it is simply connected.

Where This Session Feeds the Course

- **Depends on:** P2 (topology, connectedness, compactness of $[0, 1]$); P1 (group language, free groups preview).
- **Feeds:** S34 (profinite groups as inverse limits of deck groups); S35 (étale π_1 and Grothendieck's correspondence: the same classification theorem with étale covers); S16 ($G_{\mathbb{Q}}$ as profinite π_1 of $\text{Spec } \mathbb{Q}$); S60 (Dessins d'Enfants: monodromy of covers of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$); Act VII (topoi internalize “covers $\leftrightarrow$ π_1 -sets”).
- **Unifying thread:** a covering is a space of points over points; monodromy is evaluation along loops; the Galois correspondence is the classification of point-systems over a base.

References for this session

- [1] Hatcher, A. *Algebraic Topology*, ch.0–1 (selected: π_1 , coverings, lifting).
- [2] May, J.P. *A Concise Course in Algebraic Topology*, ch.1–3 (selected).
- [3] Munkres, J.R. *Topology*, §51–54 (fundamental group and coverings).
- [4] Szamuely, T. *Galois Groups and Fundamental Groups*, ch.2 (the Galois-correspondence viewpoint and the bridge to étale π_1).

End of Session P3 — the Prelude is complete; next: Act 0, Session S1, Rings, Ideals, and Spectra